# MCO Course Notes

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## Contents

# 1 Linear Algebra and Calculus III Background

## 1.1 Vector Spaces and $\mathbb{R}^n$

Many of the below definitions are stated in terms of general vector spaces. An important theorem to remember is the following.

**Theorem 1.1.** *A finite-dimensional vector space $V$ over scalars $\mathbb{R}$ with dimension $n$ is isomorphic to $\mathbb{R}^n$.*

*Proof.* Let $\{v_1, v_2, \ldots, v_n\}$ be a basis of $V$. Define $f : V \to \mathbb{R}^n$ via $f(a_1v_1 + a_2v_2 + \cdots + a_nv_n) = (a_1, a_2, \ldots, a_n)$. $f$ is a linear transformation and a bijection. Thus it is an isomorphism. □

**Takeaway:** All definitions and operations done in (finite-dimensional) vector spaces $V$ can be thought of as occurring in $\mathbb{R}^n$ through this standard transformation.

## 1.2 Inner Products and Norms

**Definition 1.2.** An **inner product** is an operation $\langle\rangle : V \times V \to \mathbb{R}$ for a vector space $V$ s.t. $\langle\rangle$ is linear w.r.t. $x$ when $y$ is fixed, $\langle x, y\rangle = \langle y, x\rangle$, and $\langle x, x\rangle \geq 0$ with equality iff $x = 0$.

**Definition 1.3.** A **norm** is a function $||\cdot|| : V \to \mathbb{R}$ for a vector space $V$ s.t. $||\alpha x|| = |\alpha||x|$ for all scalars $\alpha$, $||x|| \geq 0$ with equality iff $x = 0$, and $||x + y|| \leq ||x|| + ||y||$.

**Lemma 1.4** (Cauchy-Schwarz Inequality)**.** $\langle x, y\rangle \leq \sqrt{\langle x, x\rangle}\cdot\sqrt{\langle y, y\rangle}\ \ \forall x, y \in V$.

*Proof.* For any scalars $s, t$ we have $\langle sx + ty, sx + ty\rangle \geq 0$. Let's expand out and optimize $s, t$. We get $\langle x, y\rangle \leq -\frac{s}{2t}\langle x, x\rangle - \frac{t}{2s}\langle y, y\rangle$. Ultimately we want this to be $\leq \sqrt{\langle x, x\rangle} \cdot \sqrt{\langle y, y\rangle}$. We need both terms to look like $\sqrt{\langle x, x\rangle} \cdot \sqrt{\langle y, y\rangle}$ for this approach to work. Solving gives $s = 1/\langle x, x\rangle$ and $t = 1/\langle y, y\rangle$. We then handle the $x = 0$ and $y = 0$ cases separately, which is easy. □

**Fact 1.5.** $||x|| = \sqrt{\langle x, x\rangle}$ *is a norm for inner product* $\langle\rangle$.

*Proof.* It is easy to verify the scalar condition and 0 condition. Verify the triangle inequality by using the Cauchy-Schwarz inequality. □

## 1.3 Continuity and Differentiability

Let $(V, ||\cdot||)$ be a norm space.

**Definition 1.6.** A **continuous function** $f : V \to \mathbb{R}$ is a function where $\forall x, d \in V$ and $\forall \epsilon > 0$, $\exists \delta > 0$ s.t. $||f(x + \delta d) - f(x)|| < \epsilon$.

**Definition 1.7.** A **differentiable function** $f : V \to \mathbb{R}$ is a function where $\forall x \in V$, there exists a linear transformation $L : V \to \mathbb{R}$ s.t. $\forall d \in V$,

$$\lim_{h \to 0} \frac{f(x + hd) - (f(x) + L(hd))}{h} = 0$$

If we want to express the linear transformation $L$, we just need to describe the values $L(v_1), L(v_2), \ldots, L(v_n)$ for a basis $v_1, v_2, \ldots, v_n$ of $V$.

**Definition 1.8.** The **gradient vector** $\nabla f$ of differentiable function $f : V \to \mathbb{R}$ at $x \in V$ with linear transformation $L$ is the vector $\nabla f = L(v_1)v_1 + L(v_2)v_2 + \cdots + L(v_n)v_n$.

Note that the gradient vector is defined for each point $x \in V$, so more generally it is a function $\nabla f : V \to V$. Further, it allows us to express the linear transformation $L$ at each $x \in V$ as $L(d) = \langle \nabla f, d \rangle$ if we define an inner product where $\langle v_i, v_i \rangle = 1$ for each $1 \leq i \leq n$. Of course, in $\mathbb{R}^n$ we would use the standard basis $e_1, e_2, \ldots, e_n$ to describe the gradient vector.

The gradient $\nabla f(x)$ is the "direction of steepest ascent" at $x \in V$ in the following sense: for any $\epsilon > 0$, let $u = \arg\max f(x + u) : ||u|| = \epsilon$. As $\epsilon \to 0$, the maximizing $u$ will be in the same direction as $\nabla f(x)$. You can prove this by using that $f(x + u) \to f(x) + L(u) = f(x) + \langle \nabla f(x), u \rangle$. The maximum value of the inner product is achieved when $u$ is parallel to $\nabla f(x)$.

**Fact 1.9.** *For* $f(x) = ||x||^2$, $\nabla f(x) = 2x$.

As an exercise, you can prove the above fact yourself. Note that the fact that $\nabla f(x)$ is parallel to $x$ is intuitive based on the interpretation of the gradient as the direction of steepest ascent.

**Definition 1.10.** The **Jacobian matrix** $J \in \mathbb{R}^{m \times n}$ of differentiable function $f : \mathbb{R}^n \to \mathbb{R}^m$ is the matrix with row $i = (\nabla f_i)^T$, where $f_i : \mathbb{R}^n \to \mathbb{R}$ is the function for the $i$th component in the output.

The Jacobian matrix gives the best linear approximation to $f$ at $x$ in the sense that $f(x + hd) \approx f(x) + J(d) \cdot h$, with this approximation tightening as $h \to 0$ (in particular, the norm of the difference of the two sides approaches 0 as $h \to 0$).

**Definition 1.11.** Let $f : \mathbb{R}^n \to \mathbb{R}$ be a twice-differentiable function. The **Hessian matrix** $H \in \mathbb{R}^{n \times n}$ of $f$ is defined via $H_{ij} = \frac{\partial^2 f}{\partial x_i \partial x_j}$.

Note that $\frac{\partial^2 f}{\partial x_i \partial x_j} = \frac{\partial^2 f}{\partial x_j \partial x_i}$, so the Hessian is a symmetric matrix, and you do not have to worry about which partial derivative you take first. Sometimes the Hessian is written as $\nabla^2 f$, but this is poor notation, as it can also be confused with the Laplacian $\nabla^2 f$. You can tell from context what $\nabla^2 f$ means because the Hessian is an $n \times n$ matrix while the Laplacian is a function from $\mathbb{R}^n$ to $\mathbb{R}$.

**Fact 1.12.** *Let* $f : \mathbb{R}^n \to \mathbb{R}$ *be a twice-differentiable function. Then the Hessian matrix of* $f$ *is the Jacobian matrix of* $\nabla f$.

**Theorem 1.13** (Taylor Expansion)**.** *Let $f : \mathbb{R}^n \to \mathbb{R}$ be a twice-differentiable function. For all $x, d \in \mathbb{R}^n$ and $h \to 0$,*

$$f(x + hd) = f(x) + h\langle \nabla f(x), d\rangle + \frac{1}{2}h^2 d^T H(x) d + o(h^2)$$

**Fact 1.14** (Line Integral)**.** *Let $f : V \to \mathbb{R}$ be a convex differentiable function. Then $\forall x, y \in V$,*

$$f(y) = f(x) + \int_0^1 \langle \nabla f((1-\lambda)x + \lambda y), y - x\rangle d\lambda$$

You can think about this equation as making infinitely many small steps from $x$ to $y$ and using the linear approximation to $f$ for each step to get the small change in $f$. The steps are the vector $d\lambda(y - x)$.

**Fact 1.15** (Line Integral (Second Derivative))**.** *Let $f : V \to \mathbb{R}$ be a convex twice-differentiable function. Then $\forall x, y \in V$,*

$$f(y) = f(x) + \langle \nabla f(x), y - x\rangle + \int_0^1 \lambda \int_0^\lambda (y - x)^T H(\mu x + (1 - \mu)y)(y - x) d\mu d\lambda$$

The above formula is pretty gross, but you can prove it by using Fact 1.14 and Fact 1.12.

## 1.4 Eigenvalues and Eigenvectors

**Definition 1.16.** An **eigenvector** of a square matrix $A \in \mathbb{R}^{n\times n}$ is a nonzero vector $v \in \mathbb{R}^n$ s.t. $Av = \lambda v$ for some $\lambda \in \mathbb{R}$. An **eigenvalue** is a value of $\lambda$ for some eigenvector $v$.

To compute eigenvalues and eigenvectors, we want to solve the equation $Av = \lambda v$ and thus $(A - \lambda I)v = 0$ for $v \neq 0$. Thus we need $\det(A - \lambda I) = 0$. Compute the eigenvalues, and then for each eigenvalue $\lambda$, compute the eigenvectors as a basis for $\mathrm{Null}(A - \lambda I)$.

**Definition 1.17.** The **characteristic polynomial** of a square matrix $A \in \mathbb{R}^{n\times n}$ is $\det(A - \lambda I)$. The **algebraic multiplicity** of an eigenvalue $\lambda$ is its multiplicity in the characteristic polynomial. The **geometric multiplicity** of an eigenvalue $\lambda$ is the number of L.I. eigenvectors associated with $\lambda$.

**Lemma 1.18.** *Geometric multiplicity of $\lambda \leq$ Algebraic multiplicity of $\lambda$.*

**Definition 1.19.** A **diagonalizable** matrix is a square matrix $A \in \mathbb{R}^{n\times n}$ with $n$ linearly independent eigenvectors. The **diagonalization** of $A$ is $A = PDP^{-1}$, where $P \in \mathbb{R}^{n\times n}$ has columns = eigenvectors and $D \in \mathbb{R}^{n\times n}$ is a diagonal matrix with entries = eigenvalues.

Note that a diagonalizable matrix $A$ must have algebraic multiplicity = geometric multiplicity for all eigenvalues $\lambda$.

**Definition 1.20.** A **orthogonal** matrix is a square matrix $A \in \mathbb{R}^{n\times n}$ with orthonormal columns.

If $A$ is orthogonal, then $A^TA = I$ and $A^{-1}A = I$, so $A^T = A^{-1}$.

**Theorem 1.21** (Spectral Theorem). *$A \in \mathbb{R}^{n\times n}$ is symmetric $\iff$ $A$ is **orthogonally diagonalizable**, i.e. it can be diagonalized with $n$ orthonormal eigenvectors.*

If $A$ is symmetric and thus orthogonally diagonalizable, then $A = PDP^{-1} = PDP^T$ simplifies to $A = \sum_{i=1}^{n} \lambda_i v_i v_i^T$ where the $\lambda_i' s, v_i' s$ are the eigenvalues and orthonormal eigenvectors.

### 1.5 Matrix Multiplication

**Lemma 1.22.** *Let $A \in \mathbb{R}^{m\times k}$ and $B \in \mathbb{R}^{k\times n}$. Let $A_i, 1 \le i \le k$ be the matrix with the ith column of $A$ and all zeros. Let $B_i, 1 \le i \le k$ be the matrix with the ith row of $B$ (as the ith column) and all zeros. Then*

$$AB = \sum_{i=1}^{k} A_i (B_i)^T$$

**Block matrix multiplication** is a method of multiplying large matrices by viewing them as smaller matrices with matrix entries. It is very useful for reasoning about matrix multiplication, because you can reason on the block-level, and not on the entry-level.

## 2 Convex Sets and Functions

### 2.1 Basics

**Definition 2.1.** A **mathematical program** is an optimization problem of the form MP $P = \min f(x) : x \in K$ where $f : K \to \mathbb{R}$.

**Definition 2.2.** A mathematical program is

- **feasible** if it has $\ge 1$ feasible solution.
- **solvable** if it has $\ge 1$ optimal solution.
- **bounded** if there is a bound on the objective of all feasible solutions.

We can generalize our notions of open, closed, and bounded sets from $\mathbb{R}^n$ to any **norm space** $(V, ||\cdot||)$ where $||x - y||$ defines the distance between $x, y \in V$.

**Theorem 2.3.** *If $K$ is non-empty and compact, and $f$ is continuous, then $P$ is solvable.*

**Definition 2.4.** A **sublevel set** $S_\alpha = \{x \in K : f(x) \le \alpha\}$ for $\alpha \in \mathbb{R}$.

**Theorem 2.5.** *If $K$ is non-empty and closed and all sublevel sets are bounded, and $f$ is continuous, then $P$ is solvable.*

**Definition 2.6.** A **convex set** is a set $K \subseteq V$ where $V$ is a vector space and $\forall x, y \in K$ and $\lambda \in [0,1]$, $\lambda x + (1-\lambda)y \in K$.

**Definition 2.7.** A **convex function** is a function $f : K \to \mathbb{R}$ (where $K \subseteq V$ for convex set $K$ and vector space $V$) where $\forall x, y \in K$ and $\lambda \in [0,1]$,

$$f(\lambda x + (1-\lambda)y) \leq \lambda f(x) + (1-\lambda)f(y)$$

**Theorem 2.8.** *Convex sets have the following closure properties.*

1. ***Intersection** of convex sets is convex.*
2. ***Linear transformation** of a convex set is convex.*
3. ***Minkowski sum** of convex sets is convex.*
4. ***Direct product** of convex sets is convex.*

**Fact 2.9.** *$f$ is a convex function $\iff$ $Epi(f)$ is a convex set.*

**Definition 2.10.** Let $f : K \to \mathbb{R}$ be a function over domain $K \subseteq V$. $\bar{x}$ is a **global minimum** if $f(\bar{x}) \leq f(x)$ for all $x \in K$. $\bar{x}$ is a **local minimum** if $\exists \epsilon > 0$ s.t. $f(\bar{x}) \leq f(x)$ for all $x \in B(\bar{x}, \epsilon) \cap K$. $\bar{x}$ is a **strict global / local minimum** if we can replace $\leq$ with $<$.

**Note:** For local minimum to be well-defined, we need a norm space $(V, ||\cdot||)$. However according to Karan, if $V$ is a finite-dimensional vector space, the choice of norm $||\cdot||$ doesn't matter – you yield the same points as local minima.

**Theorem 2.11.** *If $f : K \to \mathbb{R}$ is a convex function and $\bar{x} \in K$ is a local minimum, then $\bar{x} \in K$ is a global minimum.*

**Lemma 2.12** (Projection Lemma)**.** *Let $K \subseteq \mathbb{R}^n$ be a convex set. Let $x \in K$, $z \notin K$, and $y = proj_K(z)$ (defined as the closest point in $K$ to $z$). Then $||x-y|| \leq ||x-z||$.*

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**Exercise 2.13.** *Prove Theorem 2.8.*

**Exercise 2.14.** *Let $S \subseteq \mathbb{R}^n$ be closed and $A \in \mathbb{R}^{m\times n}$. Is $AS$ closed?*

**Exercise 2.15.** *Let $f$ be a convex function and $\alpha \in \mathbb{R}$. Prove that $S_\alpha(f)$ is convex. Is the converse true, i.e. if $S_\alpha(f)$ is convex for all $\alpha \in \mathbb{R}$, is $f$ convex?*

**Exercise 2.16.** *Find a non-constant function $f : \mathbb{R} \to \mathbb{R}$ where all local minima are global maxima. (not a typo)*

**Exercise 2.17.** *Prove Lemma 2.12. Hint: Argue that $\angle xyz$ is a non-acute angle.*

## 2.2 NP-Hardness of Convex Optimization

I will not go through full proofs, but instead discuss the main takeaways and important definitions, because I don't think full proofs are relevant for the qualifying exam.

**Theorem 2.18.** *It is NP-hard to determine whether a point $\bar{x} \in \mathbb{R}$ is a local minimum of a degree 4 polynomial $f : \mathbb{R}^n \to \mathbb{R}$.*

The proof uses a reduction from 3-SAT. The takeaway is that in many practical situations finding a local minimum efficiently is possible, but formally speaking it is NP-hard.

**Definition 2.19.** A **homogeneous polynomial** $f : \mathbb{R}^n \to \mathbb{R}$ is a polynomial where all terms have the same degree.

**Definition 2.20.** The **homogenized polynomial** $\tilde{f} : \mathbb{R}^{n+1} \to \mathbb{R}$ of a degree $d$ polynomial $f : \mathbb{R}^n \to \mathbb{R}$ is $\tilde{f}(x, y) = y^d f(x/y)$.

You can also compute the homogenized polynomial $\tilde{f}$ by just padding each term with a $y^i, 0 \leq i \leq d$ term to make the degree $d$.

## 2.3 Descent Directions

**Definition 2.21.** Let $f : K \to \mathbb{R}$ be a function over domain $K \subseteq V$. $d \in V$ is a **descent direction** at $\bar{x} \in V$ if $\exists \bar{\epsilon} > 0$ s.t. $\forall \epsilon \in (0, \bar{\epsilon})$, $f(\bar{x} + \epsilon d) < f(\bar{x})$.

**Lemma 2.22.** *Let $f : \mathbb{R}^n \to \mathbb{R}$ be a differentiable function. If $\langle \nabla f(x), d \rangle < 0$, then $d$ is a descent direction at $x \in \mathbb{R}^n$. If $d$ is a descent direction at $x \in \mathbb{R}^n$, then $\langle \nabla f(x), d \rangle \leq 0$.*

*Proof Idea.* Use the limit definition of differentiable function. Use the fact that $L(d) = \langle \nabla f(x), d \rangle$. □

**Lemma 2.23.** *Let $f : \mathbb{R}^n \to \mathbb{R}$ be a differentiable function and $x \in \mathbb{R}^n$. Let $A =$ "$x$ is a local min", $B =$ " $\nexists$descent direction $d$ at $x$", and $C =$ "$\nabla f(x) = 0$". Then $A \implies B \implies C$, and none of these implications go in reverse.*

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**Exercise 2.24.** *Find a differentiable function $f : \mathbb{R}^n \to \mathbb{R}$ and descent direction $d$ at some $x$ s.t. $\langle \nabla f(x), d \rangle = 0$.*

**Exercise 2.25.** *Prove all cases of Lemma 2.23.*

## 2.4 Positive Semidefinite Matrices

**Definition 2.26.** A **positive semidefinite** matrix $S \in \mathbb{R}^{n \times n}$ is a square matrix which is symmetric and $\forall x \in \mathbb{R}^n, x^T S x \geq 0$.

**Theorem 2.27.** *Let $S \in \mathbb{R}^{n \times n}$ be a symmetric matrix. The following statements are equivalent.*

1. *$S$ is positive semidefinite.*

2. *All eigenvalues of $S$ are $\geq 0$.*

3. *$\exists L \in \mathbb{R}^{n \times n}$ s.t. $S = LL^T$.*

4. *$\exists v_1, v_2, \ldots, v_n \in \mathbb{R}^n$ s.t. $S_{ij} = v_i^T v_j$ for all $i, j \in [n]$.*

5. *The determinant of all principal minors of $S$ are at least $0$.*

*Proof.*

- (1) $\implies$ (2): Let $\lambda$ be an eigenvalue of $S$ and $v$ be a corresponding eigenvector (which exists because $S$ is diagonalizable). Then $Sv = \lambda v$ so $v^T S v = v^T \lambda v = \lambda v^T v$, so $\lambda = \frac{v^T S v}{v^T v} \geq 0$.

- (2) $\implies$ (3): $LL^T = \sum_{i=1}^n L_i L_i^T$. Pick $L_i = \sqrt{\lambda_i} u_i$ (in the $i$th column, $0's$ elsewhere) for each $i \in [n]$, where the $\lambda_i's, u_i's$ are eigenvalues, orthonormal eigenvectors. Then $LL^T = \sum_{i=1}^n \lambda_i u_i u_i^T = S$.

- (3) $\implies$ (4): Pick $v_i = i$th row of $L$. Note that $S = LL^T$ so $S^T = (L^T)^T L^T$ giving $S = L^T L$, so you could also pick $v_i = i$th column of $L$, i.e. the $i$th orthonormal eigenvector.

- (4) $\implies$ (1): Let $x \in \mathbb{R}^n$ be arbitrary. Then $x^T S x = \sum_{i,j} x_i x_j S_{ij} = \sum_{i,j} x_i x_j v_i^T v_j = \sum_{i,j} (x_i v_i)^T (x_j v_j) = \langle \sum x_i v_i, \sum x_i v_i \rangle \geq 0$.

Going to skip (5) $\iff$ (1) since it was mostly skipped in lecture. $\square$

## 2.5 Optimization Conditions

Unconstrained optimization is optimization over a function $f : \mathbb{R}^n \to \mathbb{R}$. Constrained optimization is optimization over a function $f : K \to \mathbb{R}$ for $K \subseteq \mathbb{R}^n$.

**Theorem 2.28** (Unconstrained Conditions)**.** *Let $f : \mathbb{R}^n \to \mathbb{R}$ be a twice-differentiable function. If $\bar{x}$ is a local min, then $\nabla f(\bar{x}) = 0$ and $\nabla^2 f(\bar{x}) \succeq 0$. If $\nabla f(\bar{x}) = 0$ and $\nabla^2 f(\bar{x}) \succ 0$, then $\bar{x}$ is a local min.*

*Proof* ($\implies$). AFSOC $\bar{x}$ is a local min but $\nabla f(\bar{x}) \neq 0$ or $\nabla^2 f(\bar{x}) \not\succeq 0$. We know $\nabla f(\bar{x}) = 0$ because of the first-order necessary condition. Thus $\nabla^2 f(\bar{x}) \not\succeq 0$, so there exists a $d \in \mathbb{R}^n$ s.t. $d^T \nabla^2 f(\bar{x}) d < 0$. Using the Taylor Expansion,

$$\begin{aligned} f(\bar{x} + hd) &= f(\bar{x}) + h\langle \nabla f(\bar{x}), d \rangle + \frac{1}{2} h^2 d^T H(\bar{x}) d + o(h^2) \\ &= f(\bar{x}) - \Theta(h^2) + o(h^2) \\ &< f(\bar{x}) \end{aligned}$$

where the final line follows as $h \to 0$, a contradiction. $\square$

*Proof* ( $\Longleftarrow$ ). Let's use Taylor Expansion again. For any unit vector $d \in \mathbb{R}^n$, $d^T H(\bar{x})d > 0$. Because the unit sphere is compact, the Extreme Value Theorem says that there exists an $m > 0$ s.t. $d^T H(\bar{x})d \geq m$ for all unit vectors $d$. Then

$$\begin{aligned} f(\bar{x} + hd) &= f(\bar{x}) + h\langle \nabla f(\bar{x}), d\rangle + \frac{1}{2}h^2 d^T H(\bar{x})d + o(h^2) \\ &\geq f(\bar{x}) + \Theta(h^2)m + o(h^2) \\ &\geq f(\bar{x}) \end{aligned}$$

Thus for sufficiently small $h$, we obtain $f(\bar{x} + hd) \geq f(\bar{x})$ for all unit vectors $d \in \mathbb{R}^n$. □

The Extreme Value Theorem is useful when you have a function that is strictly positive, and you want to show that it achieves a strictly positive minimum value (so it can't just approach zero to arbitrary precision, which would screw up our argument, because what could happen is that there is always some $d$ with $d^T H(\bar{x})d \to 0$ as $h \to 0$).

The "first-order" conditions are simply local min $\implies$ $\nabla f(\bar{x}) = 0$. The converse does not hold. This can be easily seen in the $n = 1$ case.

**Theorem 2.29** (Constrained Conditions)**.** *Let $f : K \to \mathbb{R}$ be a differentiable function for $K$ convex. If $\bar{x} \in K$ is a local min, then $\langle \nabla f(\bar{x}), x - \bar{x}\rangle \geq 0$ for all $x \in K$.*

We can write constrained second-order conditions, but checking if a matrix $A$ has $x^T Ax \geq 0$ for all $x \in K$ is NP-hard. So the second-order conditions are not that useful.

**Definition 2.30.** The **normal cone** of a convex set $K$ at $\bar{x} \in K$ is $N_k(\bar{x}) = \{d \in \mathbb{R}^n : \langle d, x - \bar{x}\rangle \leq 0 \ \forall x \in K\}$.

It's easy to verify that the normal cone is a cone. Note $d = 0$ always works. The constrained condition is simply $-\nabla f(\bar{x}) \in K$.

## 2.6 Separation Theorem

The Separation Theorem in MCO is a slight generalization of the Separating Hyperplane Theorem in LP (these names are interchangeable). I suppose the main new thing about this theorem in MCO is that we can prove a version of it.

**Theorem 2.31** (Separation Theorem)**.** *Let $A, B \subseteq \mathbb{R}^n$ be nonempty convex sets with relint$(A) \cap$ relint$(B) = \emptyset$. Then there exist $s, c \in \mathbb{R}^n$ with $s \neq 0$ s.t. $s^T x \leq c$ for all $x \in A$ and $s^T x \geq c$ for all $x \in B$. If $A$ and $B$ are closed, and $A$ or $B$ is bounded, then the separation can be made strict.*

We will prove the strict separation theorem in the case where $A \cap B = \emptyset$. This is a good proof to review because it uses a lot of facts + theorems stated before. I try to give an explanation of my thought process below.

*Proof.* Draw a 2D picture. The idea is to find the two points on $A, B$ which are closest to each other, draw the line segment between them, and define the hyperplane so that it is perpendicular to this line segment and cuts through the midpoint. Let's implement this idea.

We first need to formally show that there are two closest points on $A, B$. Assume $A$ is bounded. Define $f : A \times B \to \mathbb{R}$ via $f(a, b) = ||b - a||$. $K = A \times B$ is non-empty, closed, and $f$ is continuous. It remains to show that $f$ has bounded sublevel sets to apply Theorem 2.5. Indeed, let $\alpha \geq 0$ be arbitrary. $A$ is bounded, so it is contained in some ball of radius $r_1$. Then the subset of $B$ where $||b - a|| \leq \alpha$ for some $a \in A$ is contained in some ball of radius $2r + 2\alpha$. This is because for two points $b_1, b_2 \in B$ and two points $a_1, a_2 \in A$ with this property, $b_1$ to $a_1$ is distance $\leq \alpha$, $a_1$ to $a_2$ is distance $\leq 2r$, $a_2$ to $b_2$ is distance $\leq \alpha$, so $b_1$ to $b_2$ is distance $\leq 2r + 2\alpha$ by triangle inequality. Then you can just draw a ball of radius $2r + 2\alpha$ around $b_1$ and it will contain the subset of $B$ that is in $S_\alpha$. Thus $f$ has bounded sublevel sets, so it achieves a minimum value at some $a \in A, b \in B$.

We now define the hyperplane $s^T x = c$. Let $s = b - a$ and $c = s^T \left(\frac{a+b}{2}\right)$ so that the hyperplane cuts through the midpoint of the line segment connecting $a, b$. Simplifying gives $c = \frac{||b||^2 - ||a||^2}{2}$.

We now show $s^T x < c$ for all $x \in A$. Here is the strategy. Intuitively, we may guess that the maximum value of $s^T x$ over $x \in A$ is achieved at $x = a$, because $a$ should be the point closest to the hyperplane. The strategy becomes showing $s^T x \leq s^T a$ for all $x \in A$ and $s^T a < c$. We also want to use the fact that $A$ is convex – we haven't used that yet. Lastly, we need to use some special property of point $a$.

Clearly the special property of point $a$ is that it minimizes distance to $B$. More formally, for the function $f(x) = \min_{b \in B} ||x - b||^2$, $a$ is a minimizer. The constrained optimization conditions apply for convex $K$ (which is good – we can use that $A$ is convex) and differentiable $f$ (which is good – $f$ seems differentiable). But the issue is that $\nabla f$ is completely unclear, and this is because of the minimum over $b \in B$ which appears in the function, making it weird.

Instead, we could consider $f(x) = ||x - b||^2$. $a$ is still a minimizer, $K$ is convex, and $f$ is differentiable. $\nabla f(x) = 2(x - b)$ (see Fact 1.9). The constrained optimization conditions for local min $a$ say that $\langle \nabla f(a), x - a \rangle \geq 0$ for all $x \in A$. Simplifying,

$$\langle \nabla f(a), x - a \rangle = \langle 2(a - b), x - a \rangle = -2\langle s, x - a \rangle = -2s^T x + 2s^T a \geq 0$$

giving $s^T x \leq s^T a$ for all $x \in A$ as desired.

Lastly, we want to show $s^T a < c$. Recall $c = s^T \left(\frac{a+b}{2}\right)$. We want to show $s^T a < s^T \left(\frac{a+b}{2}\right)$ so we want to show $s^T \left(\frac{b-a}{2}\right) > 0$. This follows from the fact that $s = b - a$ and of course $s \neq 0$ because $A, B$ disjoint. Proving $s^T x > c$ for all $x \in B$ follows similarly. Done! $\square$

## 2.7 Convex Function Conditions

Let $K$ be a convex set and $f : K \to \mathbb{R}$ be differentiable.

**Proposition 2.32.** *$f$ is convex iff $\forall x, y \in K \;\; f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle$*

For proof ( $\implies$ ), proceed directly. To pick $\lambda$, and to get $\nabla f(x)$ to pop into the inequality, push $\lambda \to 0$. For proof ( $\impliedby$ ), proceed directly. Generate two inequalities using $x, \lambda x + (1 - \lambda y)$ and $\lambda x + (1 - \lambda y), y$. To deal with $\nabla f(\lambda x + (1 - \lambda)y)$, add the two inequalities (with weights $\lambda, 1 - \lambda$) to make the inner product terms cancel.

**Proposition 2.33.** *$f$ is convex iff $\forall x, y \in K \;\; \langle \nabla f(y) - \nabla f(x), y - x \rangle \geq 0$.*

For both cases, you show equivalence to the second condition in Proposition 2.32. For proof ( $\implies$ ), proceed directly. The most clever solution is to use Proposition 2.32 to generate two inequalities by flipping $x, y$ and then adding them. For proof ( $\impliedby$ ), use Fact 1.14. You want to show the integral is at least $\langle f(x), y - x \rangle$, which you can do by rearranging.

Now suppose $K$ has non-empty interior and $f$ is twice-differentiable.

**Proposition 2.34.** *$f$ is convex iff $\nabla^2 f(x) \succeq 0$ for all $x \in int(K)$.*

For both cases, you show equivalence to the second condition in Proposition 2.32. For proof ( $\implies$ ), use Taylor Expansion. For proof ( $\impliedby$ ), use Line Integral (Second Derivative).

## 2.8 Subgradients

**Definition 2.35.** Let $f : K \to \mathbb{R}$ be a function. A **subgradient** of $f$ at $x \in K$ is a vector $g \in \mathbb{R}^n$ s.t. $f(y) \geq f(x) + g^T(y - x)$ for all $y \in K$. The set of all subgradients of $f$ at $x$ is $\partial f(x)$.

Note that for a non-differentiable function $f$, the notation $\nabla f(x)$ is sometimes used to denote a subgradient of $f$ at $x$.

**Theorem 2.36.** *Let $f : K \to \mathbb{R}$ be a function for $K$ convex. If $\forall x \in K \;\; \partial f(x) \neq \emptyset$, then $f$ is convex. If $f$ is convex, then $\forall x \in int(K) \;\; \partial f(x) \neq \emptyset$.*

For proof ( $\implies$ ), proceed directly. For proof ( $\impliedby$ ), note $(f(x), x)$ is on the boundary of Epi$(f)$, and use separation theorem on $(f(x), x)$ and Epi$(f)$. I suppose the main advantage of separation theorem is that it is the only theorem which can generate inequalities for convex functions / sets (when you don't have differentiability!), other than the definition of a convex function.

---

**Exercise 2.37.** *Prove that...*

- *Non-negative sums of convex functions are convex.*
- *$f(Ax + b)$ is convex for any convex $f$.*

- $\max_y f(x, y)$ *is convex if* $\forall y, x \mapsto f(x, y)$ *is convex.*
- $g(f(x))$ *is convex if* $f, g$ *convex and* $g$ *is non-decreasing.*
- $\min_y f(x, y)$ *is convex if* $f$ *is jointly convex in* $x, y$.
- $g(x, y) = yf(x/y)$ *is convex for* $y > 0$ *if* $f$ *is convex on* $\mathbb{R}^n$.
- *Sublevel sets of convex functions are convex.*

# 3 Duality

## 3.1 The Optimization Hierarchy

**Definition 3.1.** A **mathematical program** is an optimization problem of the form MP $P = \min f(x) : x \in K$ where $f : K \to \mathbb{R}$.

**Definition 3.2.** A **convex program** is an MP where $f$ is convex.

Note that $f$ convex immediately implies $K$ convex and $K \subseteq V$ vector space.

**Definition 3.3.** A **conic program** is an MP where $f$ is convex and $K$ is the intersection of an affine subspace and a convex cone.

An affine subspace is represented by affine constraints $Ax = b$. Oftentimes $K$ is used to denote the convex cone, so a conic program takes the form $\min f(x) : Ax = b, x \in K$. We now discuss some convex cones which give rise to types of conic programs.

**Lemma 3.4.** *The following three sets are convex cones.*

1. *The* ***nonnegative orthant*** $\mathbb{R}^n_+ = \{x \in \mathbb{R}^n : x \geq 0\}$ *gives rise to LPs.*
2. *The* ***Lorentz cone*** $\mathbb{L}^{n+1} = \{(t, x) \in \mathbb{R}^{n+1} : t \geq ||x||_2\}$ *gives rise to SOCPs.*
3. *The* ***semidefinite cone*** $\mathbb{S}^{n\times n}_+ = \{S : S \succeq 0\}$ *gives rise to SDPs.*

**Definition 3.5.** A **semidefinite program** (SDP) is a conic program where $f$ is **linear**, matrices must be **symmetric**, and the cone is the **semidefinite cone**. It can be written as

$$\begin{aligned} \min_{X \in \mathbb{S}^n} \quad & \mathrm{Tr}(CX) \\ \text{s.t.} \quad & \mathrm{Tr}(A_i X) = b_i \ \ \forall i \in [m] \\ & X \succeq 0 \end{aligned}$$

There is a lot to unpack with this definition and representation. Let's first discuss the fact that this definition uses matrices instead of vectors. Recall that any finite-dimensional vector space is isomorphic to $\mathbb{R}^n$. More succinctly, any finite-dimensional vector space "is" $\mathbb{R}^n$. Thus the vector space of square $n \times n$ matrices "is" $\mathbb{R}^{n^2}$, and we can think of matrices as vectors, for example by stacking the columns into one column of length $n^2$.

For $n \times n$ matrices $A, X$, we have

$$A^T X = \begin{bmatrix} a_1^T \\ a_2^T \\ \dots \\ a_n^T \end{bmatrix} \begin{bmatrix} x_1 & x_2 & \cdots & x_n \end{bmatrix} = \begin{bmatrix} a_1^T x_1 & a_1^T x_2 & \cdots & a_1^T x_n \\ a_2^T x_1 & a_2^T x_2 & \cdots & a_2^T x_n \\ \dots & \dots & \dots & \dots \\ a_n^T x_1 & a_n^T x_2 & \dots & a_n^T x_n \end{bmatrix}$$

and thus $\mathrm{Tr}(A^T X) = a_1^T x_1 + a_2^T x_2 + \cdots + a_n^T x_n$. The point is that trace allows us to write any linear combination of the entries of $X$! This is why $\mathrm{Tr}(CX)$ and $\mathrm{Tr}(A_i X)$ is used in the objective and linear constraints. Note we often write $\langle A, X \rangle = \mathrm{Tr}(A^T X)$.

Lastly, we need to explain how to handle $X \in \mathbb{S}^n$. The constraint that a matrix is symmetric can be written as linear constraints $X_{ij} = X_{ji}$ for all $i, j \in [n]$. Thus to show an SDP is a conic program, these constraints would appear in the linear constraints $Ax = b$.

**Definition 3.6. SDP Engineering** gives techniques for transforming "complicated" SDP constraints into the standard SDP format. There are three main techniques:

1. **Wiring:** Given the constraint $M \succeq 0$ where $M$'s cells consists of affine terms $a^T z - b$, we can replace each cell with a variable $x_{ij}$, and then "wire" $x_{ij} = a^T z - b$ as a linear constraint. The resulting matrix $M$ consists of only single variables.

2. **Packing:** Given a variable $x$ which does not appear in $X$, we can replace $x \leftarrow z^+ - z^-$ where $z^+, z^- \geq 0$. Then we can update $X \leftarrow \begin{bmatrix} X & 0 & 0 \\ 0 & z^+ & 0 \\ 0 & 0 & z^- \end{bmatrix}$. Now $z^+, z^-$ appear in the variable matrix, and the constraint that the new matrix is PSD is equivalent to $X \succeq 0, z^+, z^- \geq 0$.

3. **Combining:** Given multiple PSD constraints $X_1, X_2, \dots, X_m \succeq 0$, we can build a block diagonal matrix $X = \mathrm{diag}(X_1, X_2, \dots, X_m)$ and require $X \succeq 0$.

We now introduce more classes of optimization problems, and at the end we show the hierarchy among them.

**Definition 3.7.** A **second-order cone program** (SOCP) is a conic program where $f$ is **linear** and constraints are **norm** and **linear**. It can be written as

$$\begin{aligned} &\min\ f^T x \\ &\text{s.t. } ||A_i x + b_i||_2 \leq c_i^T x + d_i \ \ \forall i \in [m] \\ &\qquad Fx = g \end{aligned}$$

**Definition 3.8.** A **convex quadratically-constrained quadratic program** (convex QCQP) is a conic program where $f$ is **quadratic**, constraints are **quadratic**, and matrices are **symmetric PSD**. It can be written as

$$\begin{aligned} &\min\ x^TQx + q^Tx \\ &\text{s.t.}\ \ x^TQ_ix + q_i^Tx + d_i \leq 0 && \forall i \in [m] \\ &\qquad Q, Q_i \succeq 0 && \forall i \in [m] \end{aligned}$$

**Definition 3.9.** A **convex quadratic program** (convex QP) is a conic program where $f$ is **quadratic**, constraints are **linear**, and the objective matrix is **symmetric PSD**. It can be written as

$$\begin{aligned} &\min\ x^TQx + q^Tx \\ &\text{s.t.}\ \ Ax = b \\ &\qquad Q \succeq 0 \end{aligned}$$

**Definition 3.10.** A **linear program** (LP) is a conic program where $f$ is **linear** and constraints are **linear**. It can be written as

$$\begin{aligned} &\min\ c^Tx \\ &\text{s.t.}\ \ Ax = b \end{aligned}$$

We now show the optimization hierarchy. First, we state a helpful lemma.

**Lemma 3.11** (Schur Complement Lemma)**.** *Let* $A \succ 0$*. Then* $X = \begin{bmatrix} A & B \\ B^T & C \end{bmatrix} \succeq 0$ *iff* $C - B^TA^{-1}B \succeq 0$*.*

**Theorem 3.12** (The Optimization Hierarchy)**.** *LP* $\subseteq$ *convex QP* $\subseteq$ *convex QCQP* $\subseteq$ *SOCP* $\subseteq$ *SDP.*

*Proof.*

- **SOCP $\subseteq$ SDP:** Via Lemma 3.11, $||x||_2 \leq t$ is equivalent to $\begin{bmatrix} tI & x \\ x^T & t \end{bmatrix} \succeq 0$. Thus each norm constraint can be written as an SDP constraint. We can use wiring and packing to clean each SDP constraint into the standard format, and then combining to combine the cleaned SDP constraints together.

- **convex QCQP $\subseteq$ SOCP:** Because $Q$ is symmetric PSD, $Q = \sum_i \lambda_i u_i u_i^T$ for $\lambda_i \geq 0$ via Theorem 2.27, and so $Q^{1/2} = \sum_i \lambda_i u_i u_i^T$. Thus the objective is $x^TQx + 2q^Tx = ||Q^{1/2}x + Q^{-1/2}q||_2^2 - q^TQ^{-1}q$. We can replace the objective as min $t$ and then add the constraint $||Q^{1/2}x + Q^{-1/2}q||_2 \leq t$. The true objective value is simply $t^2 - q^TQ^{-1}q$. We can do the same for all of the quadratic constraints, writing them as $||Q_i^{1/2}x + Q_i^{1/2}q_i||_2 \leq \sqrt{q_i^TQ_i^{-1}q_i - d_i}$. Note that if $q_i^TQ_i^{-1}q_i - d_i < 0$, then the constraint is infeasible.

- **convex QP $\subseteq$ convex QCQP:** Set $Q_i = 0$ for all $i \in [m]$.
- **LP $\subseteq$ convex QP:** Set $Q = 0$.

$\square$

---

**Exercise 3.13.** *Give an SDP with infinitely many extreme points.*

**Exercise 3.14.** *Give an SDP which is bounded below but not solvable.*

**Exercise 3.15.** *Write* $\max c^T x : Ax \leq b$ *as an SDP.*

**Exercise 3.16.** *Show that SOCP constraints can be written as the intersection of an affine subspace and the Lorentz cone.*

**Exercise 3.17.** *Show that any convex program can be written as a conic program for some convex cone* $K$*.*

## 3.2 BQPs and MAXCUT

**Definition 3.18.** A **binary quadratic program** (BQP) is a mathematical program that can be written as

$$\begin{aligned} &\max\ x^T A x \\ &\text{s.t. } x \in \{-1, 1\}^n \\ &\quad\ \ A \succeq 0 \end{aligned}$$

I don't actually know if this is the general format of a BQP, but it's the format used for this application. Note a BQP is a **non-convex** optimization problem, as the constraint $x \in \{-1, 1\}^n$ is clearly non-convex. It can be relaxed to an SDP in the following way: $\max \operatorname{Tr}(A^T x x^T)$ s.t. $x_i^2 = 1$ for all $i \in [n]$, which is $\max \operatorname{Tr}(A^T X)$ s.t. $X_{ii} = 1$ for all $i \in [n]$, $X \succeq 0$ and $X$ is rank 1. Then to relax it, throw away the "$X$ is rank 1" constraint to get the SDP. Clearly SDP $\geq$ BQP.

**Theorem 3.19.** *BQP* $\geq \frac{2}{\pi}$ *SDP.*

I'll skip the proof, but I'll show some useful lemmas.

**Lemma 3.20.** *If* $A \succeq 0$ *and* $B \succeq 0$*, then* $\langle A, B \rangle \geq 0$*.*

*Proof.* Write $A = \sum_i \lambda_i u_i u_i^T$. Then $\langle A, B \rangle = \sum_i \lambda_i \langle u_i u_i^T, B \rangle = \sum_i \lambda_i u_i^T B u_i \geq 0$. $\square$

**Lemma 3.21.** *If* $A \succeq 0$ *and* $B \succeq 0$*, then* $A \odot B \succeq 0$*.*

*Proof.* $A = \sum_i \lambda_i v_i v_i^T$ and $B = \sum_j \mu_j u_j u_j^T$, so $A \odot B = \sum_{ij} \lambda_i \mu_j (v_i v_i^T) \odot (u_j u_j^T) = \sum_{ij} \lambda_i u_j (v_i \odot u_j)(v_i \odot u_j)^T$. $\lambda_i u_j \geq 0$ and the matrices are PSD, so $A \odot B$ is a nonnegative linear combination of PSD matrices and thus is PSD. $\square$

$\odot$ above is element-wise product. The connection to MAXCUT is that we can formulate MAXCUT using a BQP, then relax it to an SDP, and round the SDP solution to get a BQP solution. The proof of Theorem 3.19 uses this rounding procedure.

## 3.3 Dynamical Systems

Let $f : \mathbb{R}^n \to \mathbb{R}^n$ be a function and $x_0, x_1, x_2, \ldots$ be a sequence where $x_{t+1} = f(x_t)$. We will consider $f(x_t) = Ax_t$ where $A$ is diagonalizable.

**Definition 3.22.** $A$ is **stable** if $x_t \to 0$ as $t \to \infty$ for all $x_0 \in \mathbb{R}^n$.

**Theorem 3.23.** *$A$ is stable iff $\rho(A) = \max_i |\lambda_i(A)| < 1$.*

**Definition 3.24.** The **spectral norm** of a matrix $A$ is $||A||_2 = \max_{||x||_2=1} ||Ax||_2$. Equivalently, $||A||_2$ is the square root of the maximum eigenvalue of $A^T A$.

**Theorem 3.25.** *$A$ is stable iff $\exists P \succeq I$ s.t. $P \succ A^T PA$.*

*Proof* ($\implies$). Skip □

*Proof* ($\impliedby$). Potential-based method. Let $V(x) = x^T Px$ where $P \succeq I$ and $cP \succeq A^T PA$ for $c \in [0, 1)$. Then $V(0) = 0, V(x) > ||x||_2^2 > 0$ for all $x \neq 0$, and $V(Ax) = x^T APAx \leq cV(x)$. Thus $V(A^t x) \to 0$ as $t \to \infty$, so $x_t \to 0$. □

Note that $\{P \succeq I, cP \succeq A^T PA\}$ is an SDP-feasibility program and thus the minimum $c$ can be found via binary search in $[0, 1)$.

Let's now consider a dynamical system $f(x_t) = Ax_t + Bu_t$ where $u_1, u_2, \ldots$ are inputs. Let's call this dynamical system $(A, B)$.

**Definition 3.26.** $(A, B)$ is **stabilizable** if for all $x_0 \in \mathbb{R}^n$, there exist $u_1, u_2, \dots \in \mathbb{R}^m$ s.t. $x_t \to 0$ as $t \to \infty$.

**Theorem 3.27.** *$(A, B)$ is stabilizable iff there exists a matrix $K$ s.t. $\rho(A + BK) < 1$.*

In the above theorem, we use $u_t = Kx_t$ as inputs. The takeaway is that using Theorem 3.25, we can determine stabilizability of $(A, B)$ using SDPs.

## 3.4 Conic Duality

We will use the notation $Ax \geq_K b$ to mean $Ax - b \in K$ for a cone $K$, because this will show more clearly how conic duality is a generalization of LP duality. We will consider the primal program $P = \min_x \{c^T x : Ax \geq_K b\}$.

**Definition 3.28.** The **dual cone** of a cone $K \subseteq \mathbb{R}^n$ is $K_* = \{y \in \mathbb{R}^n : y^T x \geq 0 \ \forall x \in \mathbb{R}^n\}$.

### 3.4.1 Conic Duality as Lower Bound Certificates

Let's think about attempting to lower-bound $c^T x$ over $Ax \geq_K b$. We do so by taking a linear combination of the rows of $A$, which is given by some $y \in \mathbb{R}^m$. Here is what the derivation looks like:

$$c^T x = y^T Ax = y^T (Ax - b) + b^T y \geq b^T y$$

The initial equality forces $A^T y = c$. The inequality forces $y \in K_*$. Thus we get the dual problem $\max_y \{b^T y : A^T y = c, y \geq_{K_*} 0\}$.

### 3.4.2 Conic Duality as a Two-Player Game

**Theorem 3.29** (Max-Min Inequality). *Let $f : X \times Y \to \mathbb{R}$ be a function. Then $\max_{x\in X} \min_{y\in Y} f(x, y) \le \min_{y\in Y} \max_{x\in X} f(x, y)$.*

Conceptually, the max-min inequality states that there is a "second-player advantage" to the two-player game of minimizing and maximizing $f(x, y)$. Then we have

$$
\begin{aligned}
\min_x \{c^T x : Ax \ge_K b\} &= \min_x \max_{y\in K_*} (c^T x + y^T (b - Ax)) \\
&\ge \max_{y\in K_*} \min_x ((c - A^T y)^T x + b^T y) \\
&= \max_{y\in K_*} \{b^T y : A^T y = c\} \\
&= \max_y \{b^T y : A^T y = c, y \ge_{K_*} 0\}
\end{aligned}
$$

We can understand this derivation as a two-player game between player $x$ and player $y$. The first line follows from an assessment of player $x$'s strategy. If player $x$ chooses $Ax \ge_K b \implies Ax - b \in K$, then $y^T (b - Ax) \le 0$ for all $y \in K_*$, and so player $y$ will choose $y = 0$, recovering the original optimization problem. If player $x$ chooses $Ax \not\ge_K b$, then $Ax - b \notin K$. Because $(K_*)_* = K$, we have $Ax - b \notin (K_*)_*$, so there exists a $y \in K_*$ s.t. $y^T (Ax - b) < 0$ giving $y^T (b - Ax) > 0$. Then player $y$ will select $ty$ and push $t \to \infty$, pushing the quantity to $\infty$. Thus player $x$ will certainly ensure that $Ax \ge_K b$. The second line follows by the max-min inequality. The third line follows by an assessment of player $y$'s strategy. If $A^T y \ne c$, then player $x$ can push the quantity to $-\infty$, so player $y$ will certainly ensure that $A^T y = c$.

### 3.4.3 Conic Duality: Strong Duality

We have established **weak duality** between primal and dual conic programs. Does strong duality hold, and if so, under what conditions?

**Lemma 3.30.** *Let $K \subseteq \mathbb{R}^n$ be nonempty. Then*

1. *$K_*$ is a closed convex cone.*
2. *$int(K) \ne \emptyset \implies K_*$ is pointed.*
3. *$K$ is closed convex pointed cone $\implies int(K_*) \ne \emptyset$.*
4. *$K$ is closed convex cone $\implies (K_*)_* = K$.*

Let's call $K \subseteq \mathbb{R}^n$ a **good cone** if it is a closed convex pointed cone with nonempty interior.

**Corollary 3.31.** *$K$ is a good cone iff $K_*$ is a good cone.*

**Theorem 3.32** (Strong Duality)**.** *If $K$ is a good cone, $P$ is bounded below, and $P$ is strictly feasible ($\exists x\ Ax >_K b$), then strong duality holds. Similarly if $K$ is a good cone, $D$ is bounded above, and $D$ is strictly feasible ($\exists y\ y >_{K_*} 0$), then strong duality holds.*

*Proof Idea.* Let $P^*$ be the optimal primal objective, and find a dual feasible $y^*$ s.t. $b^T y^* \leq P^*$. To do so, define $V = \{Ax - b : c^T x \leq P^*\}$. Argue $V \cap \text{int}(K) = \emptyset$, and use separation theorem to get a $y \in \mathbb{R}^m \backslash \{0\}$ s.t. $\max_{x \in V} y^T x \leq \min_{x \in K} y^T x$. You can argue $y$ is dual feasible and $b^T y \leq P^*$. $\square$

## 3.5 $S$-Lemma

Consider a non-convex QCQP with a single constraint: $\min_x\{q_a(x) : q_b(x) \geq 0\}$ s.t. $q_a(x) = x^T Q_a x + q_a^T x + c_a$ and $q_b(x) = x^T Q_b x + q_b^T x + c_b$. It turns out that this non-convex QCQP is efficiently solvable, despite being non-convex. There is a way to recast this as an SDP.

An important lemma is the $S$-Lemma, which is essentially Homogeneous Farkas' Lemma for quadratic constraints.

**Lemma 3.33** ($S$-Lemma)**.** *Suppose $\exists x$ s.t. $q_b(x) > 0$. If $q_b(x) \geq 0 \implies q_a(x) \geq 0$, then $\exists \lambda \geq 0$ s.t. $q_a(x) \geq \lambda q_b(x)$ for all $x$.*

The reverse implication is obvious. The $S$-Lemma is interesting because it's not immediately obvious that if $q_a(x) \geq 0$ is a consequence of $q_b(x) \geq 0$, then such a certificate $\lambda \geq 0$ exists. I will skip the proof, and instead state some useful lemmas.

**Lemma 3.34.** *Let $A, B$ be symmetric matrices. If $Tr(A) \geq 0 > Tr(B)$, then $\exists x$ s.t. $x^T A x \geq 0$ and $x^T B x < 0$.*

*Proof Idea.* Use the fact that $\mathbb{E}[x^T M x] = \text{Tr}(M)$ over $x \in \{-1, 1\}^n$ for any matrix $M$. Then use a probabilistic argument. $\square$

**Lemma 3.35.** *Suppose $\exists x$ s.t. $x^T A x > 0$. If $x^T A x \geq 0 \implies x^T B x \geq 0$, then $\exists \lambda \geq 0$ s.t. $B \succeq \lambda A$.*

*Proof Idea.* SDP Duality. Let $\text{OPT} = \min_x\{x^T B x : x^T A x \geq 0\}$. Considering $\text{OPT}' = \min_x\{x^T B x : x^T A x \geq 0, x^T x = n\}$ gives a more useful SDP dual (I am not exactly sure why, but I'm guessing if you take the SDP dual of OPT, it gives something useless). Recast as $\text{OPT}' = \min_x\{\text{Tr}(BX) : \text{Tr}(AX) \geq 0, \text{Tr}(X) = n, X \text{ is rank } 1\}$ where $X = xx^T$. Drop the rank 1 constraint to get an SDP, and take the SDP dual to prove the existence of $\lambda$. $\square$

**Note:** $x^T A x = \text{Tr}(AX)$ for $X = xx^T$. This is because you can write $A = \begin{bmatrix} a_1^T \\ a_2^T \\ \dots \\ a_n^T \end{bmatrix}$ and $xx^T = \begin{bmatrix} (xx^T)_1 & (xx^T)_2 & \dots & (xx^T)_n \end{bmatrix} = \begin{bmatrix} x_1 x & x_2 x & \dots & x_n x \end{bmatrix}$.
Then you get

$$\mathrm{Tr}(AX) = \sum_i a_i^T (xx^T)_i = \sum_i a_i^T x_i x = \sum_i x_i (a_i^T x) = x^T A x$$

This is useful for converting $x^T Ax$ objectives and constraints into semidefinite objectives and constraints. Upon performing this substitution you do generate a constraint that $X$ is rank 1 because $X = xx^T$, but you can drop this constraint to get an SDP.

# 4 Algorithms for Convex Optimization

**WARNING:** At this point in my studying I got low on time, so I have some of the main theorems below, but this section is definitely not as extensive as the previous sections. At certain points where I say "I believe..." I would check for yourself because I did not extensively check that what I was saying was correct.

## 4.1 Ellipsoid Method

Let $f : K \to \mathbb{R}$ be a convex function where $K$ contains a ball of radius $r$ and is contained in a ball of radius $R$. The ellipsoid method solves the convex optimization problem $\min\{f(x) : x \in K\}$.

**Definition 4.1.** A **separation oracle** for a convex and compact set $K$ is an oracle that given $x \in K$, returns 1) $x \in K$ or 2) a hyperplane which strictly separates $x$ from $K$.

**Ellipsoid Algorithm:** Let $\mathcal{E}_1$ be the ball of radius $R$ enclosing $K$. For $t = 1, 2, \ldots, T$, let $x_t$ be the center of $\mathcal{E}_t$. Call a separation oracle on $x_t$. If $x_t \in K$, find an ellipsoid $\mathcal{E}_{t+1}$ enclosing $\mathcal{E}_t \cap \{x : f(x) \leq f(x_t)\}$. Else $x_t \notin K$, so find an ellipsoid $\mathcal{E}_{t+1}$ enclosing $\mathcal{E}_t$ and $K$'s half-space of the returned hyperplane. Upon completion of the for loop, output the latest point $x_t \in K$.

**Separation Algorithm:** If $K$ is polyhedral, then we can check $x_t \in K$ by checking the constraints. Else $x_t \notin K$, which means some inequality is violated, and this gives the separating hyperplane.

**Lemma 4.2** (Volume Reduction Lemma)**.** *For any ellipsoid $\mathcal{E}$ with center $x_0 \in \mathbb{R}^n$ and vector $v \in \mathbb{R}^n$ defining the hyperplane $v^T(x - x_0) = 0$ cutting through $x_0$, we can efficiently construct an ellipsoid $\mathcal{E}'$ enclosing $\mathcal{E} \cap \{x \in \mathbb{R}^n : v^T(x - x_0) \leq 0\}$, s.t. $vol(\mathcal{E}') \leq vol(\mathcal{E}) \cdot \exp\left(-\frac{1}{2(n+1)}\right)$.*

**Lemma 4.3.** *Suppose the convex optimization problem is an LP, i.e. $f(x) = c^T x$ and $K$ is polyhedral. Let $F = \max_{x,y \in K} c^T(x - y)$. Then for any $\epsilon > 0$, the ellipsoid algorithm returns $x^*_\epsilon \in K$ s.t. $c^T x^*_\epsilon \leq c^T x + \epsilon$ after $T = O\left(n^2 \log \frac{FR}{\epsilon r}\right)$ rounds.*

*Proof.* Let $X^*_\epsilon = (1 - \epsilon)x^* + \epsilon K \subseteq K$. $X^*_\epsilon$ has volume $\Theta((\epsilon r)^n)$. After $O(n)$ rounds, the volume of the ellipsoid drops by a constant factor, and thus after

$O\left(n \log(R/(\epsilon r))^n\right) = O\left(n^2 \log \frac{R}{r\epsilon}\right)$ rounds, the volume of the ellipsoid drops below $\text{vol}(X_\epsilon^*)$. Thus there exists some $t, x \in X_\epsilon^*$ s.t. $x \in \mathcal{E}_t$ and $x \notin \mathcal{E}_{t+1}$. $x$ must have gotten cut off due to having low objective, and note that its objective is at most $c^T x^* + \epsilon F$. Thus by running the algorithm for $O\left(n^2 \log \frac{FR}{r\epsilon}\right)$ rounds, we can obtain an $\epsilon$-optimal solution. □

**Note:** An ellipsoid $\mathcal{E} \subseteq \mathbb{R}^n$ is given by $\mathcal{E} = \{x \in \mathbb{R}^n : (x-x_0)^T A(x-x_0) \leq 1\}$ where $x_0 \in \mathbb{R}^n$ and $A \succ 0$.

## 4.2 Gradient Descent for $L$-Smooth Functions

**Definition 4.4.** $f : \mathbb{R}^n \to \mathbb{R}$ is **$L$-smooth** if it is differentiable and $||\nabla f(x) - \nabla f(y)|| \leq L \cdot ||x - y||$ for all $x, y \in \mathbb{R}^n$.

**Lemma 4.5.** *$f$ is $L$-smooth iff $f(y) \leq f(x) + \nabla f(x)^T(y-x) + \frac{L}{2}||y-x||_2^2$ for all $x, y \in \mathbb{R}^n$.*

Suppose $f$ is twice-differentiable.

**Lemma 4.6.** *$f$ is $L$-smooth iff $\nabla^2 f(x) \preceq LI$ for all $x \in \mathbb{R}^n$.*

We now discuss the gradient descent algorithm for $L$-smooth functions.
**Input:** An $L$-smooth function $f : \mathbb{R}^n \to \mathbb{R}$ and point $x_0 \in \mathbb{R}^n$.
**Output:** An $\epsilon$-critical point $x \in \mathbb{R}^n$, i.e. a point where $||\nabla f(x)||_2 \leq \epsilon$.
**Algorithm:** Compute $x_{t+1} \leftarrow x_t - \frac{1}{L}\nabla f(x_t)$ for $T = O\left(\frac{L(f(x_0)-f(x^*))}{\epsilon^2}\right)$ rounds. Output an $\epsilon$-critical point $x_t$ once found.
**Analysis:** Use Lemma 4.5 to show $f(x_{t+1}) \leq f(x_t) - \frac{1}{2L}||\nabla f(x_t)||_2^2$ for all $t$. Then sum over $t = 1, 2, \ldots, T$ to upper bound $\frac{1}{T}\sum_{t=1}^{T} ||\nabla f(x_t)||_2^2$ and use an average argument.
**Note:** Critical point $\not\Rightarrow$ minimizer of $f(x)$, but it can be a good proxy.

## 4.3 Gradient Descent for $\mu$-Strongly Convex Functions

**Definition 4.7.** $f : \mathbb{R}^n \to \mathbb{R}$ is $\mu$-strongly convex for $\mu \geq 0$ if

$$f(\lambda x + (1-\lambda)y) \leq \lambda f(x) + (1-\lambda)f(y) - \frac{\mu}{2}\lambda(1-\lambda)||y-x||_2^2$$

for all $x, y \in \mathbb{R}^n$ and $\lambda \in [0, 1]$.

Suppose $f : \mathbb{R}^n \to \mathbb{R}$ is differentiable.

**Lemma 4.8.** *$f$ is $\mu$-strongly convex iff $f(y) \geq f(x) + \nabla f(x)^T(y-x) + \frac{\mu}{2}||y-x||_2^2$ for all $x, y \in \mathbb{R}^n$.*

**Note:** $\mu = 0$ gives Proposition 2.32.
Now suppose $f$ is twice-differentiable.

**Lemma 4.9.** *$f$ is $\mu$-strongly convex iff $\nabla^2 f(x) \succeq \mu I$ for all $x \in \mathbb{R}^n$.*

**Note:** $\mu = 0$ gives Proposition 2.34.

**Theorem 4.10.** *Let $f : \mathbb{R}^n \to \mathbb{R}$ be an $L$-smooth $\mu$-strongly convex function. Then for gradient descent $x_{t+1} \leftarrow x_t - \frac{1}{L}\nabla f(x_t)$ we have*

$$f(x_t) - f(x^*) \leq \left(1 - \frac{\mu}{L}\right)^t (f(x_0) - f(x^*))$$

This implies that $\epsilon$-optimality requires $T = O\left(\frac{L}{\mu}\log\frac{f(x_0)-f(x^*)}{\epsilon}\right)$ rounds. Note that $\gamma = \mu/L$ is sometimes called the **condition number** of an $L$-smooth $\mu$-strongly convex function. $\gamma \in [0, 1]$ and a high $\gamma$ implies fast convergence.

## 4.4 Gradient Descent for More Functions

**Theorem 4.11.** *Let $f : \mathbb{R}^n \to \mathbb{R}$ be a differentiable $L$-smooth convex function. Then for gradient descent $x_{t+1} \leftarrow x_t - \frac{1}{L}\nabla f(x_t)$ we have*

$$f(x_t) - f(x^*) \leq O\left(\frac{L||x_0 - x^*||^2}{t}\right)$$

One way to get a simplified but suboptimal (by extra $\log t$ factor) proof of the above theorem is to add a controlled amount of strong convexity to the function, i.e. use $g(x) = f(x) + \frac{\alpha}{2}||x - x_0||^2$. $g$ is $\alpha$-strongly convex. You then optimize $\alpha$.

**Definition 4.12.** $f : \mathbb{R}^n \to \mathbb{R}$ is $G$-Lipschitz continuous if $||f(x) - f(y)|| \leq G||x - y||$ for all $x, y \in \mathbb{R}^n$.

**Lemma 4.13.** *Let $f : \mathbb{R}^n \to \mathbb{R}$ be a convex function. Then $f$ is $G$-Lipschitz continuous iff $||\nabla f(x)||_2 \leq G$.*

**Lemma 4.14.** *Suppose $f : \mathbb{R}^n \to \mathbb{R}$ is $G$-Lipschitz continuous and $\alpha$-strongly convex. Define $S : \mathbb{R}^n \to \mathbb{R}$ via $S(x) = \mathbb{E}_{||u||_2=1}[f(x + \delta u)]$ for $\delta > 0$. Then $S$ is $\alpha$-strongly convex, $nG/\delta$-smooth, and $||S(x) - f(x)|| \leq \delta G$ for all $x \in \mathbb{R}^n$.*

The above lemma allow us to optimize non-smooth Lipschitz strongly convex functions $f$ by performing gradient descent on their smoothed function $S$. Optimizing over $\delta$ gives the following theorem.

**Theorem 4.15.** *Let $f : \mathbb{R}^n \to \mathbb{R}$ be a $G$-Lipschitz continuous and $\alpha$-strongly convex function. Then there exists $\delta > 0$ s.t. for gradient descent $x_{t+1} \leftarrow x_t - \frac{1}{L}\nabla S(x_t)$ where $S$ is the $\delta$-smoothed function of $f$, we have*

$$f(x_t) - f(x^*) \leq O\left(\frac{nG^2 \log t}{\alpha t}\right)$$

## 4.5 Online Gradient Descent

In **online convex optimization**, the initial input consists of a convex set $K \subseteq \mathbb{R}^n$ and (finite) family of convex functions $\mathcal{F}$. At step $t$, the online player chooses $x_t \in K$, the adversary chooses $f_t \in \mathcal{F}$, and the online player suffers cost $f_t(x_t)$. The online player's goal is to minimize **regret**, defined as

$$\text{Regret}(T) = \sum_{t=1}^{T} f_t(x_t) - \min_{x \in K} \sum_{t=1}^{T} f_t(x)$$

i.e. the online player attempts to perform as closely as possible to the optimal "single-point" strategy.

**Algorithm:** Compute $x_{t+1} \leftarrow \text{proj}_K(x_t - \eta_t \nabla f_t(x_t))$.

**Theorem 4.16.** *Let $D = \max_{x,y \in K} ||x - y||$ be the diameter of $K$ and $G$ be the maximum Lipschitz constant of any function in $\mathcal{F}$. Then Online Gradient Descent with step sizes $\eta_t = \frac{D}{G\sqrt{t}}$ guarantees Regret$(T) \leq \frac{3}{2} GD\sqrt{T}$.*

*Proof Idea.* We know $f_t(x^*) - f_t(x_t) \geq \nabla f_t(x_t)^T(x^* - x_t)$ and thus $f_t(x_t) - f_t(x^*) \leq \nabla f_t(x_t)^T(x_t - x^*)$ by convexity. We can upper bound the right hand side by noticing that such a term arises in the computation of $||x_t - \nabla f_t(x_t)^T - x^*||_2^2 \geq ||x_{t+1} - x^*||_2^2$ (the inequality follows by Lemma 2.12). We can then sum over $t = 1, 2, \ldots, T$ to get Regret$(T)$ on the left hand side, and a telescoping upper bound on the right side (in terms of $\eta_t$). Optimizing $\eta_t$ yields $\eta_t = \frac{D}{G\sqrt{t}}$ and the desired result. $\square$

## 4.6 Online Mirror Descent

Mirror Descent is a generalization of Gradient Descent that uses the **Bregman divergence**.

**Definition 4.17.** The **Bregman divergence** $D_h : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$ of a continuously-differentiable strictly convex function $h : \mathbb{R}^n \to \mathbb{R}$ is given by $D(x, y) = h(x) - h(y) - \langle \nabla h(y), x - y \rangle$.

Note that Taylor expansion gives $h(x) = h(y) + \langle \nabla h(y), x - y \rangle + D_h(x, y)$.

**Algorithm (Mirror Descent):** Mirror Descent starts with a convex function $f : K \to \mathbb{R}$, an $\alpha$-strongly convex function $h : \mathbb{R}^n \to \mathbb{R}$, and a point $x_0 \in K$. On each step, let $\theta_t \leftarrow \nabla h(x_t)$, update $\theta_{t+1} \leftarrow \theta_t - \eta_t \nabla f(x_t)$, let $x'_{t+1} \leftarrow (\nabla h)^{-1}(\theta_{t+1})$, and finally let $x_{t+1} \leftarrow \arg\min_{x \in K} D_h(x, x'_{t+1})$.

**Algorithm (Online Mirror Descent):** Same as Mirror Descent, but we update according to $\nabla f_t(x_t)$ on each step.

## 4.7 Lower Bounds for Convex Optimization

We consider the following convex optimization algorithm framework: at each time $t$, the algorithm has a history $h_t = (x_1, g_1, \ldots, x_t, g_t)$ of points and subgradients (where $g_i \in \partial f(x_i)$ for $i = 1, 2, \ldots, t$). The point $x_{t+1}$ is computed deterministically from $h_t$. Importantly, the algorithm only has **gradient oracle access** to the function, meaning it can only query the function's gradient, and I believe it can only do one gradient query per $x_t$.

**Theorem 4.18.** *There exists a convex L-Lipschitz function $f : \mathbb{R}^n \to \mathbb{R}$ s.t. for any algorithm in the framework and $t \leq n$,*

$$\min_{1\leq s\leq t} f(x_s) - \min_{||x||\leq R} f(x) \geq \Omega(RL/\sqrt{t})$$

*Further, there exists an $\alpha$-strongly convex L-Lipschitz function $f : \mathbb{R}^n \to \mathbb{R}$ s.t. for any algorithm in the framework, $t \leq n$, and $R = \frac{L}{2\alpha}$, we have*

$$\min_{1\leq s\leq t} f(x_s) - \min_{||x||\leq R} f(x) \geq \Omega(RL/t)$$

For both parts, the function used is $f(x) = \gamma \max_{1\leq i\leq t} x(i) + \frac{\alpha}{2}||x||_2^2$. Then $\gamma, \alpha$ are optimized to obtain the lower bounds. I didn't explicitly mention it anywhere, but gradient descent has $O(1/\sqrt{t})$ and $O(1/t)$ convergence rates in these settings, matching the lower bounds.

**Theorem 4.19.** *There exists a convex $\beta$-smooth function $f : \mathbb{R}^n \to \mathbb{R}$ s.t. for any algorithm in the framework and $t \leq (n-1)/2$,*

$$\min_{1\leq s\leq t} f(x_s) - \min_{||x||\leq R} f(x) \geq \Omega\left(\frac{\beta||x_1 - x^*||_2^2}{t^2}\right)$$

Note Theorem 4.11 shows that gradient descent achieves $O\left(\frac{\beta||x_1-x^*||_2^2}{t}\right)$ convergence in this setting. Thus there is a factor of $t$ gap between the upper bound achieved by gradient descent and the lower bound required by any algorithm in the framework.

## 4.8 Omitted Topics

Unfortunately, due to time constraints, I have omitted bandit convex optimization, multi-armed bandits problem, accelerated gradient descent via linear coupling, and the heavy-ball method for quadratic optimization.